Introduktion til Systemudvikling

- General Introduktion 1. semester

Important things

- Nametags
- Register attendence

The way the Semester is run

- Many different topics – a ballance between getting an overview but also producing a result and reflections on the process.
- The module are based on dialog, practical experiments, groupwork and report writting. A module where you have to attend.
- It is an introdutionary module, where the ability to analyses, design, implement and reflect is evaluated.

Literature

- Reading material is to be fund on fronter
- Craig Larman (2004). Applying UML and Patterns : An Introduction to Object-Oriented Analysis and Design and Iterative Development (3rd Edition). Prentice-Hall.
- The internet / ChatGPT

Overview

Læringsmål Systemudvikling

Systemudvikling = 1. og 2. Semester

Link til studieordningen

https://katalog.kea.dk/course/3050142/2024-2025

Process of the exam

- The examination takes place Jun 2026
- The exam is group based + individual and without preparation
- The 30 minutes for the group and 30 minutes for each student. The examination is based on the project report and takes the process related task each student has been responsible for into account. The project report provide a base for the examiners to start the questioning. Though the quality of the report is not determining the grading, it of course gives an indication.
- The grading is according to the 12-grade scale. The grade expresses the evaluation of the report, the reflection on the process-oriented task by the student and the oral examination. The project itself is not graded.
- 5 – 10 minutes for grading and entrance of the next student.

Exam Project
second semester

- An important part of the project take in consideration a specific organization which the students follow and writes their project for.
- In the very beginning of the project the students are placed in a group.
- Groups are not final before every student are part of one.
- Discuss and agree upon aspiration level
- Discuss and agree upon working method and degree of delegation

My classes

- Be carefull of absence
- The way I teach: repetition, self study in groups, presentation of new stuff, you work with it

The study – how we work

- You are (hopefully)
  - Adults
  - Mature
  - Have chosen to be here voluntarily
  - Here to learn

- So we expect you to be
  - Active
  - Prepared
  - Tolerant and respectful (both to the teacher and other students!)

The study – how we work

- In class, it is OK to
  - Ask questions
  - Solve exercises together
  - Ask others for help
  - Ask the teacher for help
  - Bring and consume a ”neutral drink” (bottle of water, soft drink, etc.)
  - Put your cell phone on silent mode…

The study – how we work

- In class, it is <u>not</u> OK to
  - Talking on your cell phone
  - Talk loudly about irrelevant matters
  - Behave noisily in general
  - Disrespect others, if they e.g. ask what you consider a trivial question
  - Eat while seated at the PC
  - Sleep in class…

The study – how we work

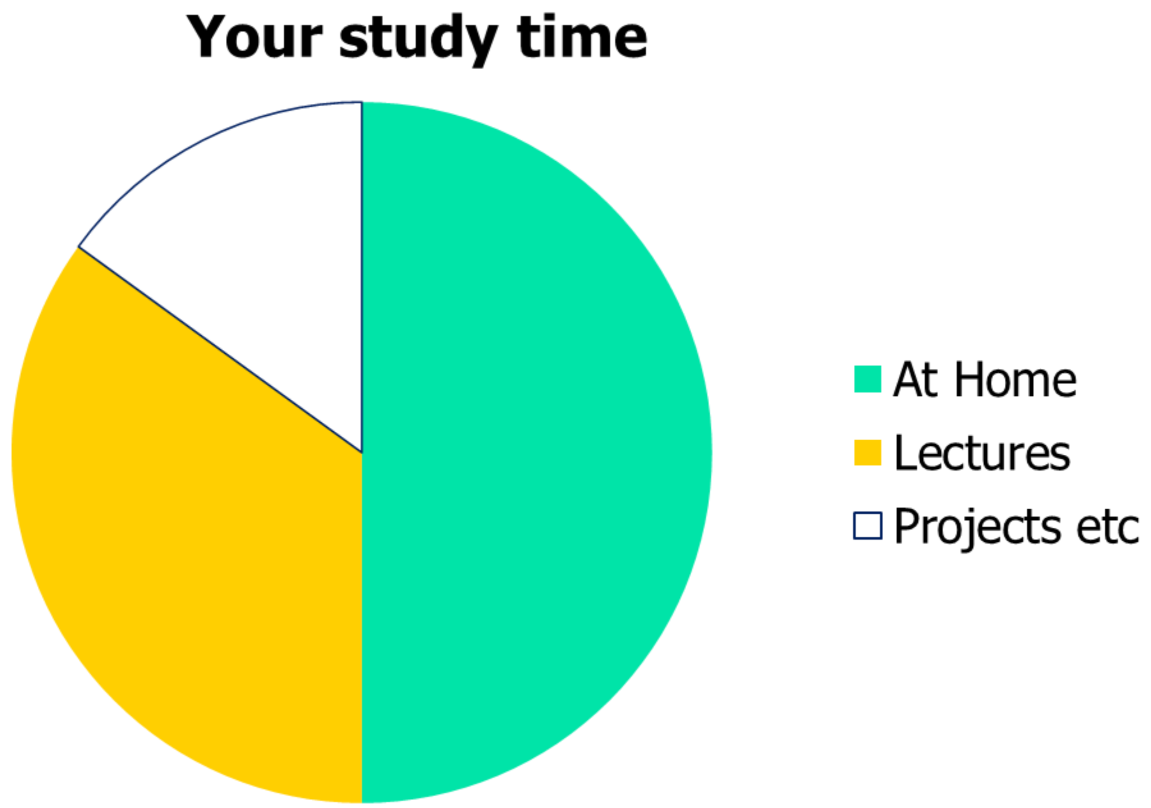
- How much am I supposed to work?
- A bit of math:
  - Each semester is 30 ECTS
  - 1 ECTS is about 28 hours of work
  - So, each semester represents 840 hours of work
  - A semester is about 20 weeks, including the exam period
  - **About 42 hours of work each week**

The study – how we work

The study – how we work

- The majority of work is done at home or in studygroups!
  - Study the curriculum proactively
  - Explore the tools
  - Practice!
- Having e.g. a full-time job along with the study will be <u>very</u> difficult

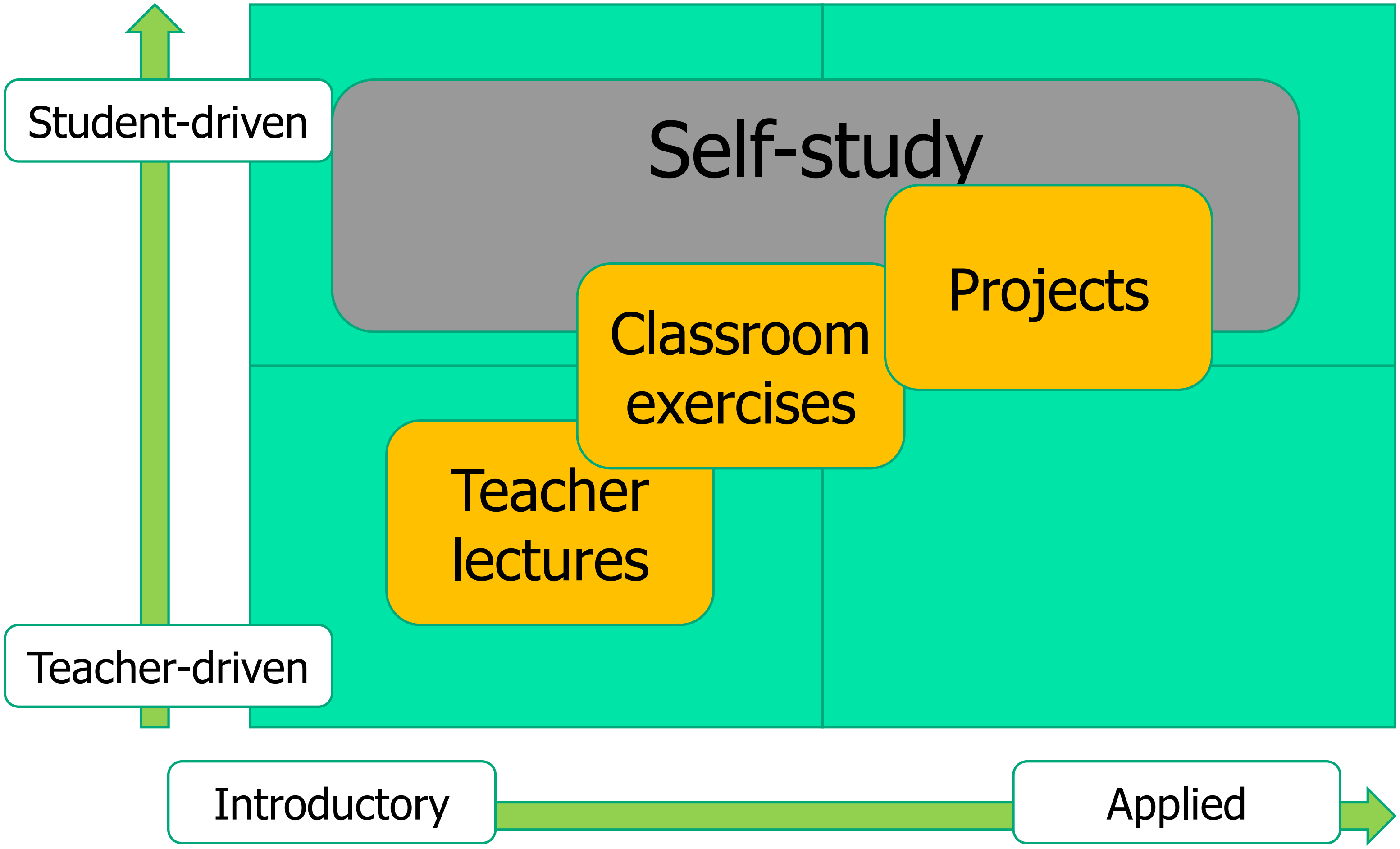
The study – how we work



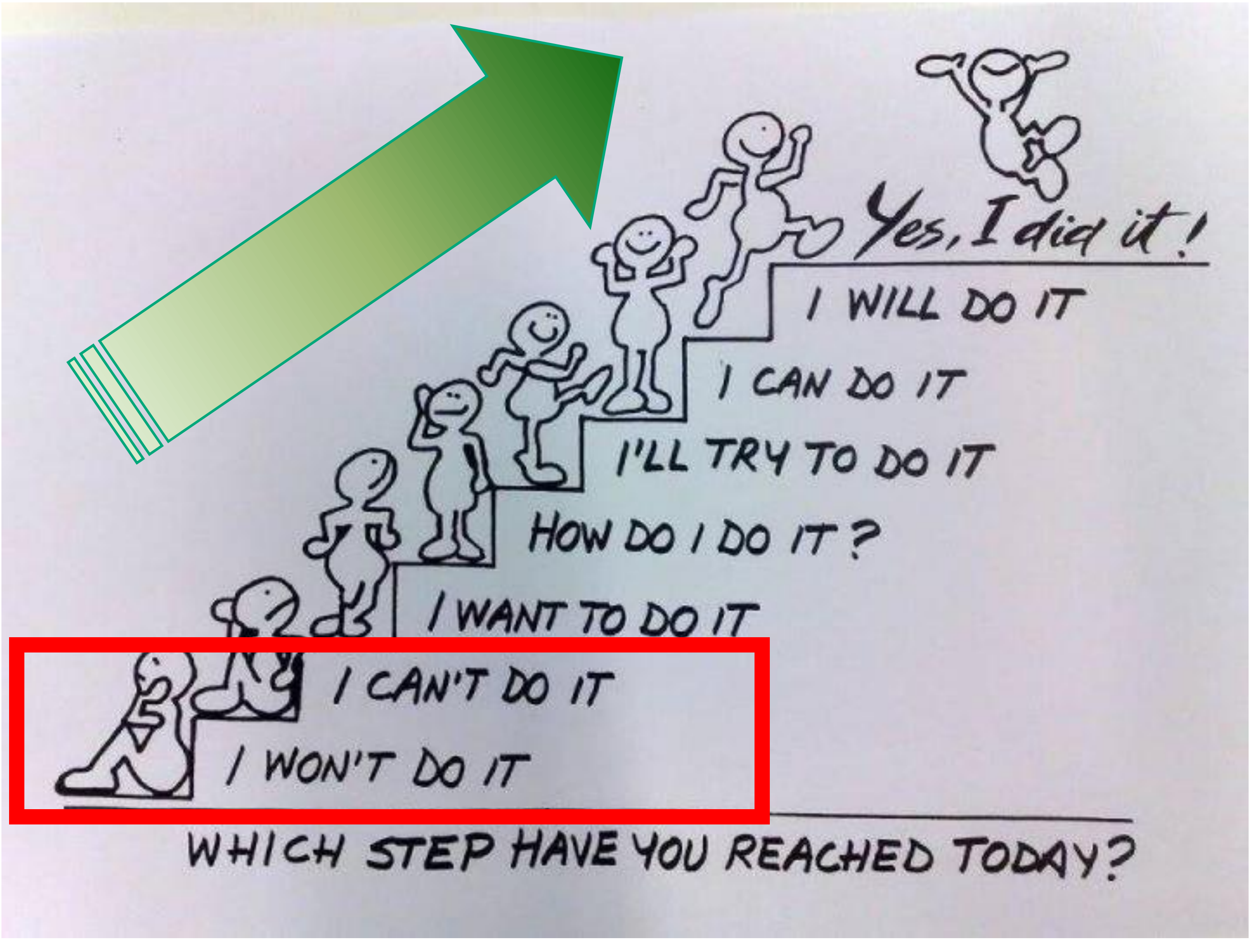
The study – how we work

- Main learning elements
  - Teacher lectures
  - Classroom exercises
  - Projects
  - Self study

The study – how we work



The study – how we work



In software design You will learn….

Thinking in objects

Requirements analysis

Design of solution

- Patterns (GRASP)
  - Principles and guidelines
- System Architecture

Processmodels

Waterfall / SCRUM

UML

What is Software?

Software is:

(1) **instructions** (computer programs) that when executed provide desired features, function, and performance

(2) **data structures** that enable the programs to adequately manipulate information and

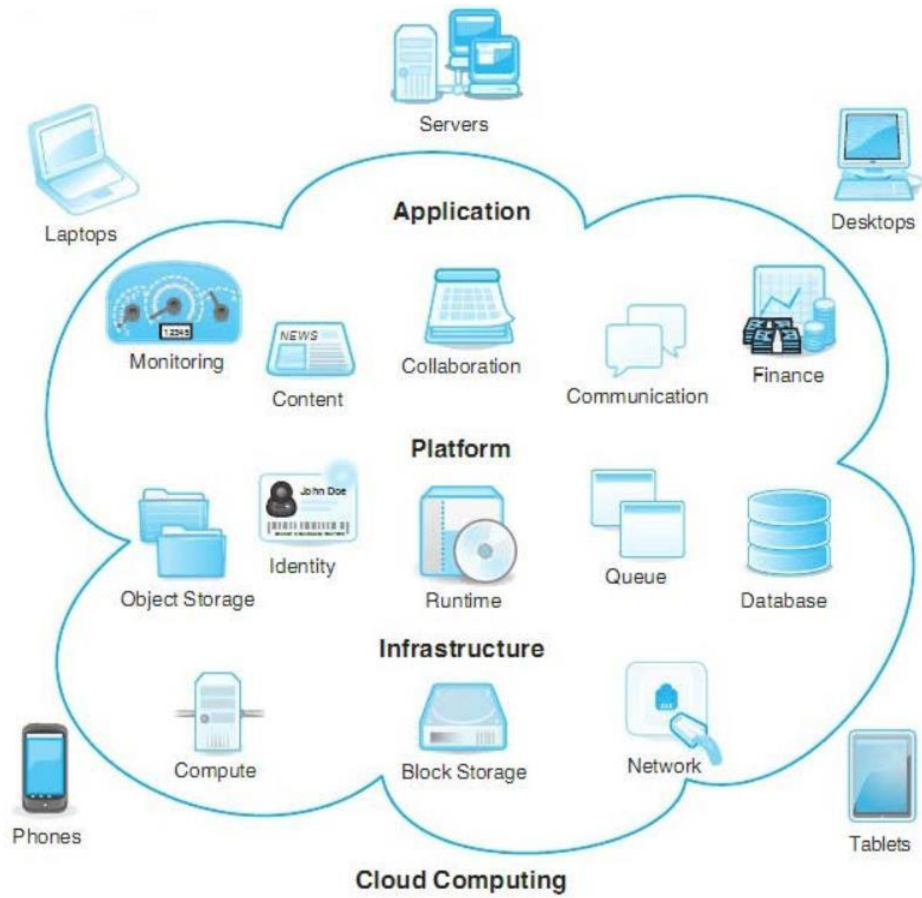
(3) **documentation** that describes the operation and use of the programs.

Software application domains

- System software
- Application software.
- Engineering/Scientific software.
- Embedded software.
- Product-line software.
- Web/Mobile applications.
- AI software (robotics, neural nets, game playing).

The changing nature of software

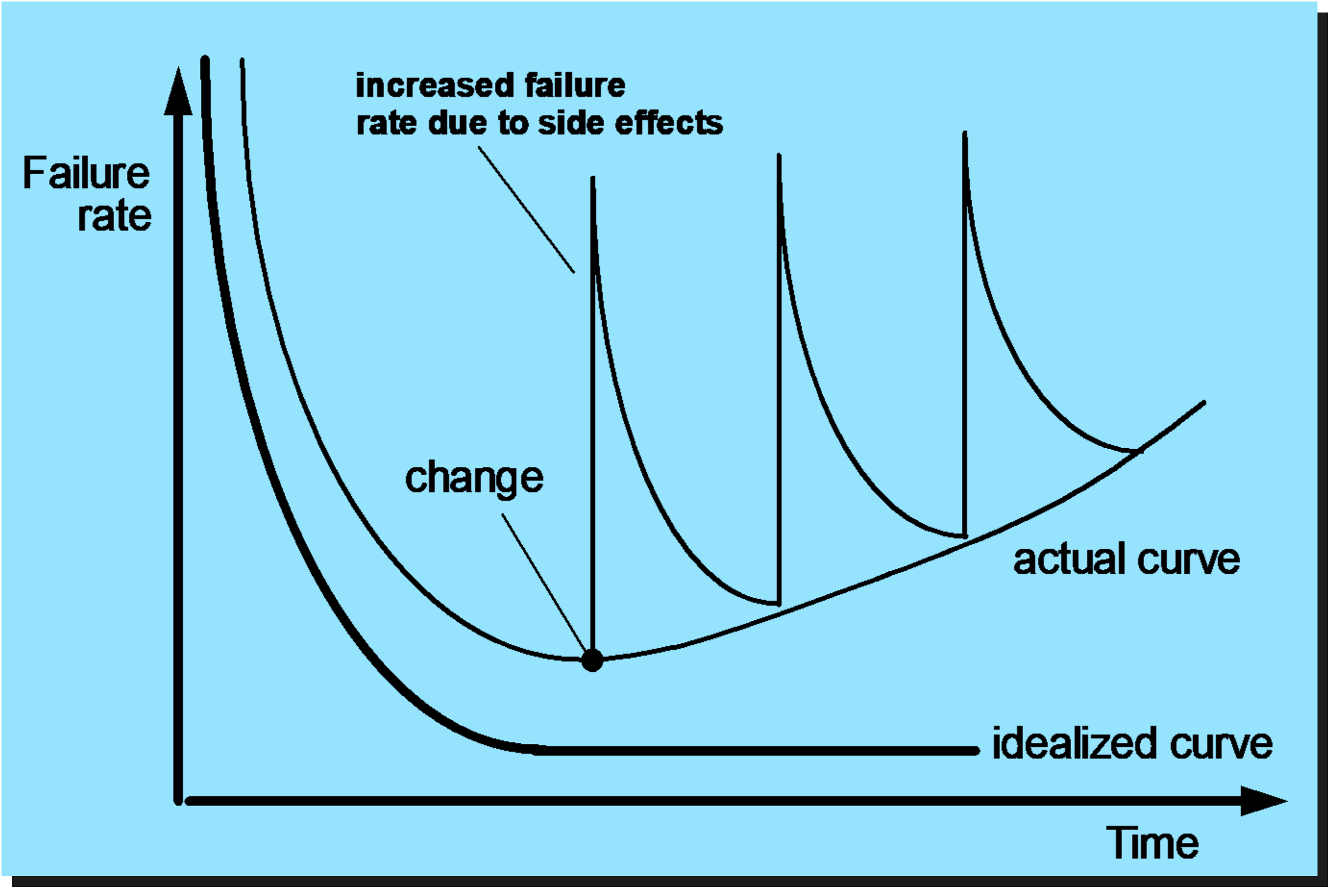
- WebApps
- Mobile Applications
- Cloud Computing
- Product Line Software



Cloud computing logical architecture

Wear vs. Deterioration

Software is developed , it is not manufactured in the classical sense. Software doesn't "wear out."



Legacy software

as time passes…

- Why must software change?
- Software must **be adapted** to meet the needs of **new computing environments or technology**
- Software must **be enhanced** to implement **new business requirements**
- Software must **be extended** to **make it interoperable with other**
- more modern systems or databases
- Software must **be re-architected** to **make it viable within a network environment**

…must be re-engineered ☺

Elements of an Information System

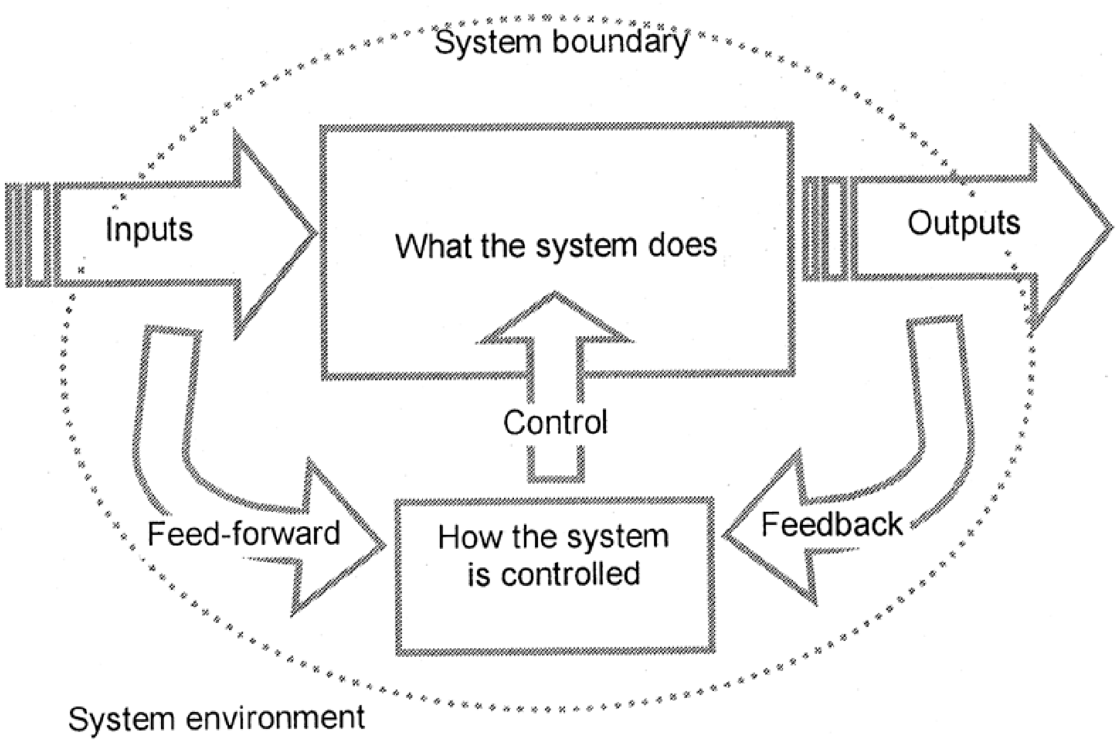
- Every IS has:
  - A human activity that needs information
  - Some stored data
  - An input method for entering data
  - Some process that turns the data into information
  - An output method for representing information

All useful systems *transform* their inputs into useful outputs

For IS, both inputs and outputs are typically information

This *transformation* is the whole reason for building and operating the system

Characteristics of Systems

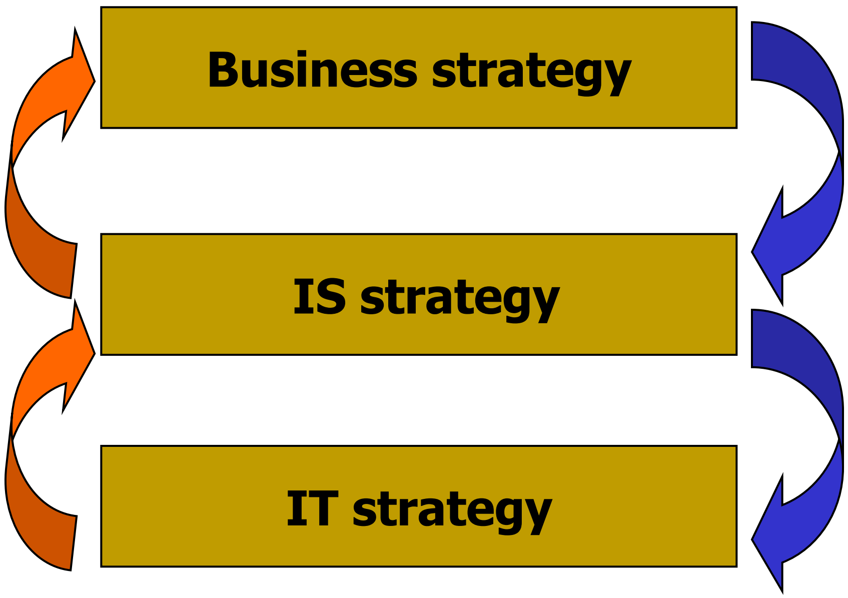


- Output from one system can be input to another system. The two systems share a part of their interface.
- A system can be part of a bigger system then it is a subsystem.

Information Systems in organizations

- An IS must support the organization to achieve its business goals.

Where can an IS help?



Which systems can support the business strategy?

Which tecnology is needed to implement the IS strategy?

Which tecnologies exists?

What is Software design?

Software design is about:

- Identifying requirements for the new system
- Which effects do we want the new system to have on the organization?
- How do we make sure that the new system meets the requirements?

Problems in Systems Development

projects fails for many reasons

The Main Players

Endusers:

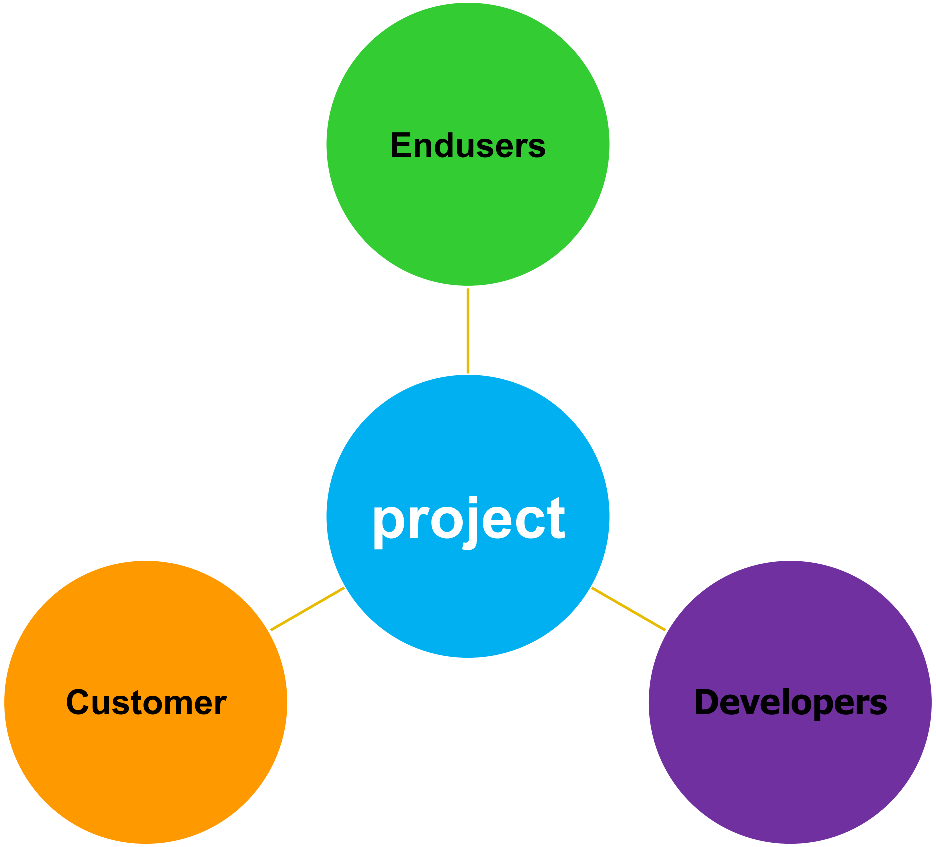
- Those who will benefit from the system’s outputs, directly or indirectly

Customer:

- Those who commission the project, pay for it or have the power to halt it.

Developers:

- Those who will produce the software



Every player sees the systems development process from their own perspective

Enduser View

Typical concerns include:

- **A system that is promised but not delivered**
- **A system that is difficult to use**
  - Poor interface design
  - Poor response times
  - Unhelpful help messages (“wrong date format-try again”)
- **A system that doesn’t meet its users’ needs**
  - Lack of functionality, useless functionality etc.

Customer View

Typical concerns include:

- **Projects that overspend their budget (may no longer have a net benefit)**
- **Systems that are delivered too late**
- **Badly managed projects**
- **Systems that have become irrelevant**
  - *Business conditions have changed*
  - *Requirements has changed.*
  - *External events.*

Developer View

- IS developers sometimes have a difficult time
  - Budget and time constraints often conflict with doing the job properly
  - Users and owners may not know how to ask for what they really want
  - Technologies and development approaches constantly change

Communication is hard

Common Glosery

Term	Description
Bibliometrics	The quantitative study of literatures as they are reflected in bibliographies
Citation network	Network of citation relations between items (e.g., publications, authors or journals)
CiteSpaceII	Software tool developed by Chen for detecting and visualizing emerging trends and transient patterns in scientific literature
CitNetExplorer	Software tool developed by Van Eck and Waltman for visualizing and analyzing citation networks of scientific publications
Mapping	Positioning of a subset of the publications in a citation network (usually

Why things go wrong

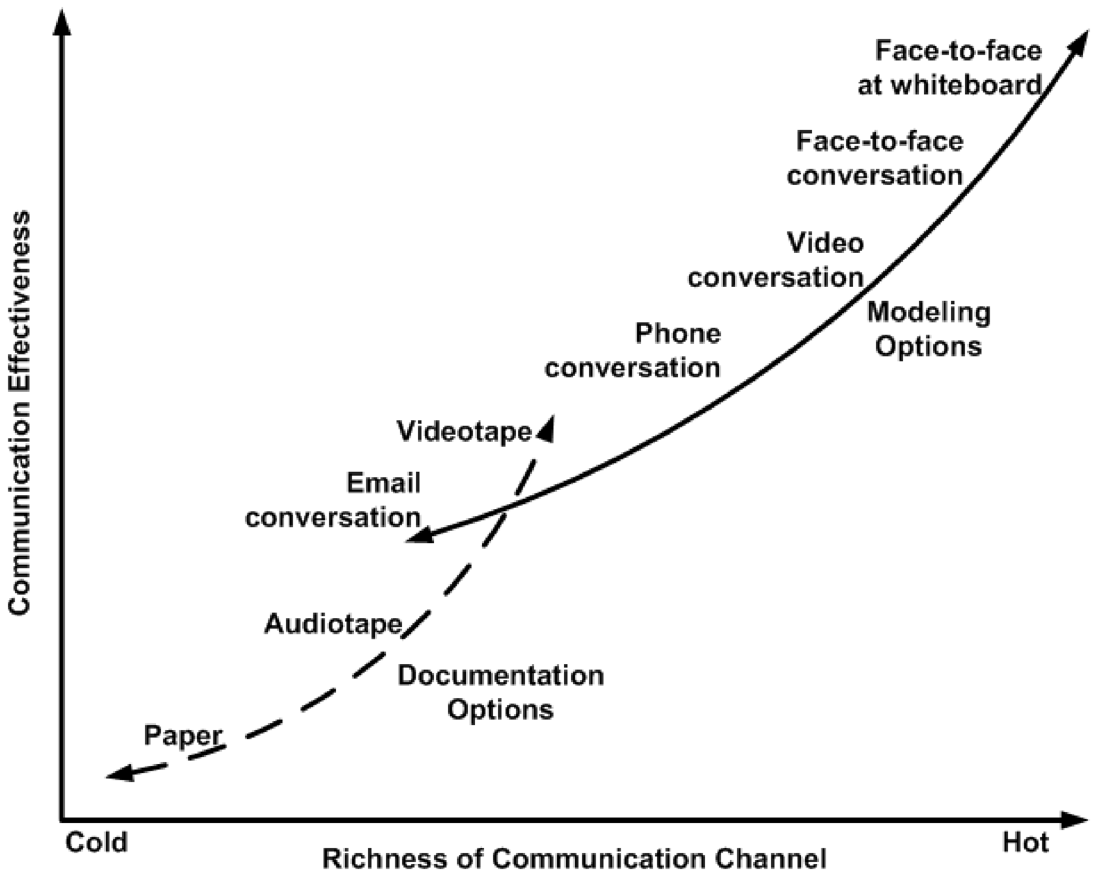
- Quality problems
- Project management problems

The underlying problem is the complexity of the project!

All systems development models are about:

- Understanding the situation/problem
- Control the complexity
- Find a solution

Effectiveness of Communication Channels



Ping-pong-pair

How:

Answer question 1 take turns with the student next to you.

Same procedure for question 2

Question 1: **which points so far did you find most important?**

Question 2: **What was not quite clear?**

You got 15 minutes!

Controlling Complexity

<u>We can control complexity by:</u>

- Subdividing the development process into a number of subtasks.

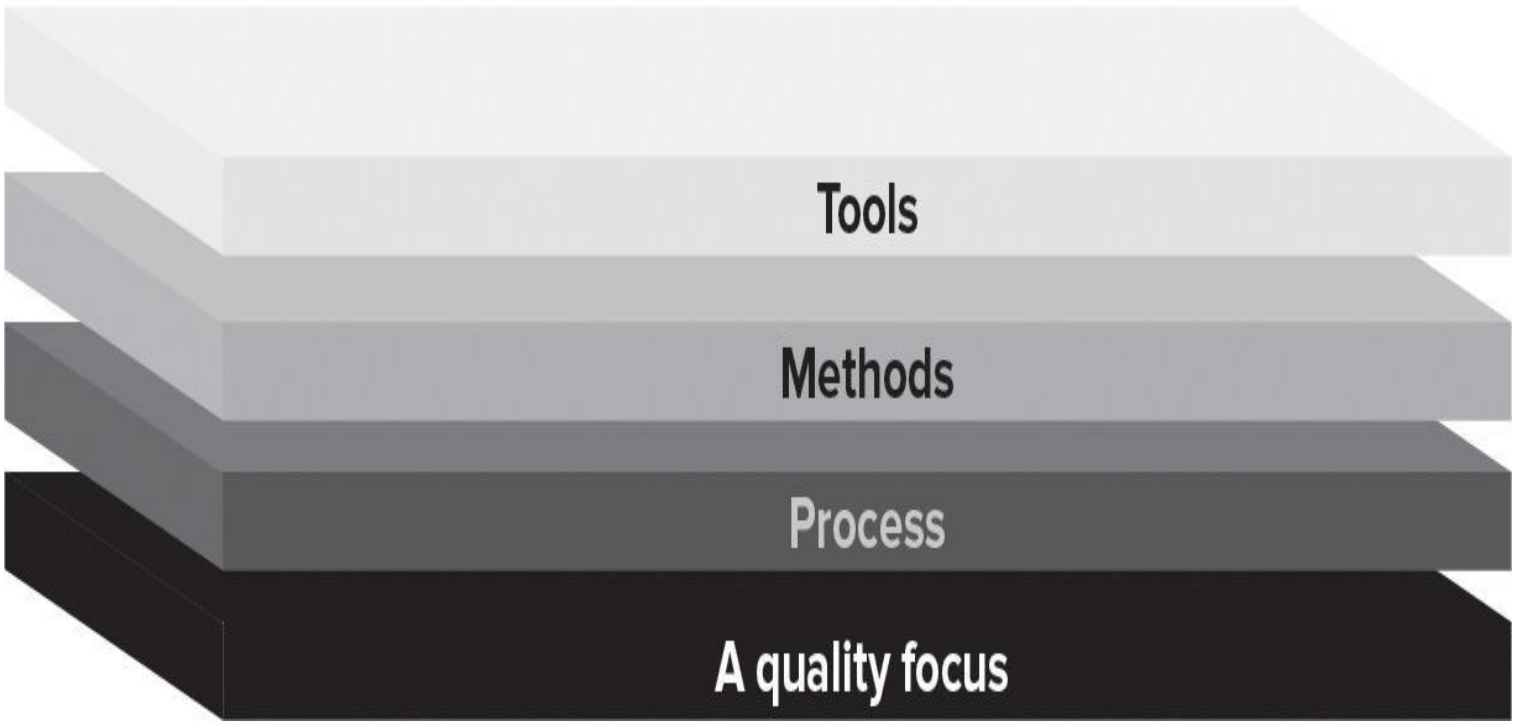
- For each task special knowledge and techniques are used

- Project management will be enhanced because activities associated with each task will be easier to define and control

General order of activities

- Gather information/requirements. Register them in UML
- Analyse the requirements
- Design preliminary solutions
- Develop software solution
- Test the solution
- Implement the solution

Software Engineering layers



Process framework activities

- Communication.
- Planning.
- Modeling.
  - Analysis of requirements.
  - Design.
- Construction:
  - Code generation.
  - Testing.
- Deployment.

Umbrella activities

- Software project tracking and control
- Risk management
- Software quality assurance
- Technical reviews
- Measurement
- Software configuration management
- Reusability management
- Work product preparation and production

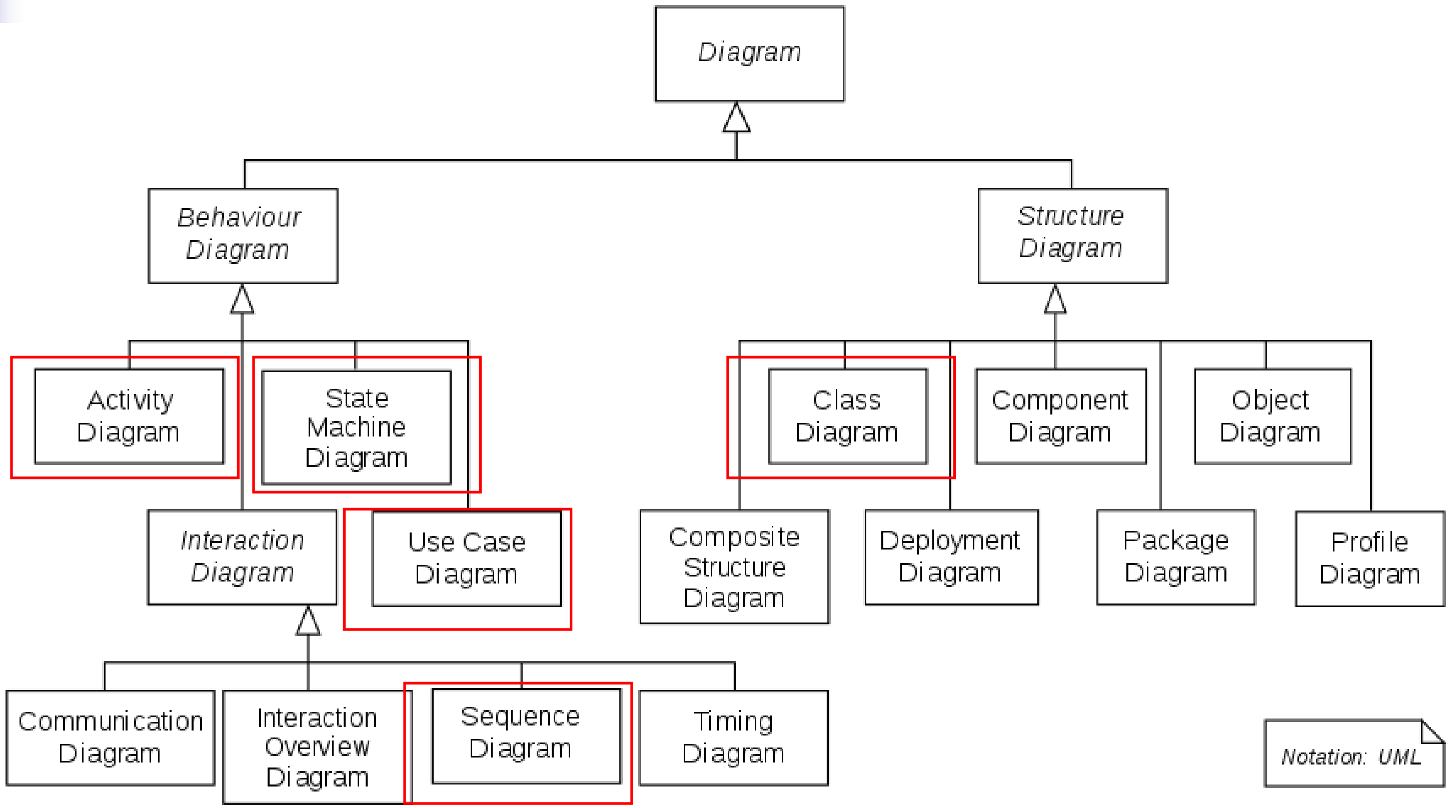
Unified Modeling Language

- A visual language for drawing diagrams related to software.

UML is not :

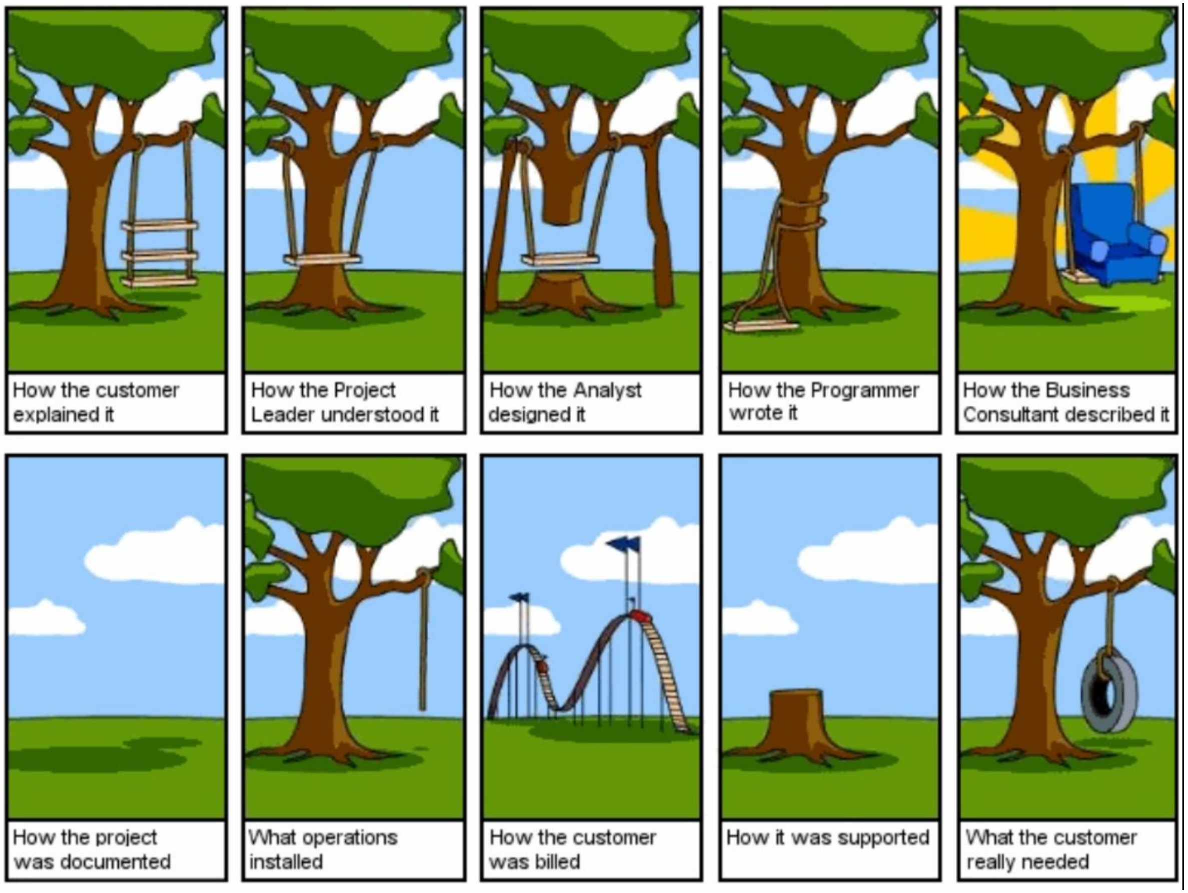
- A process or method
- Object-oriented Analysis and Design
- Design guidelines

UML: Diagrams overview



https://en.wikipedia.org/wiki/Unified_Modeling_Language

Why UML?



https://www.slideshare.net/MikelRaj/ooad-unit-i-uml-diagrams

Summary

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Opgave

- Lav Intro til softwareudvikling i grupper af 4 personer

Install Visual Paradigme

Se hvordan på Fronter i mappen Visual Paradigme

Prøv herefter at lave diagrammer i VP